



Capacity or Volume

In our everyday life, we measure the quantities of various liquids such as petrol, oil, milk, etc.

The quantity of a liquid that a container or vessel can hold is called its **capacity** or **volume**.

Thus, the capacity of a container tells us how much of a liquid it can hold.

Units of Measuring Capacity

The standard unit of capacity or volume is **litre**.

We measure the quantities of liquids such as milk, petrol, oil and water in litres.

We denote 'litres' by L.

Millilitre is the smaller unit of capacity or volume. Cold drinks, medicinal liquids, shampoos, perfumes etc. in small bottles are measured in millilitres.

We denote 'millilitres' by mL.

Liquids like oil are measured with the help of standard sized vessels shown below.



Liquids like milk and water are measured with the help of standard sized vessels, shown below.



The relation between litre and millilitre is given by:

$$1\text{ L} = 1000\text{ mL}$$

Conversion of Litres into Millilitres and Vice Versa

Rule 1: To convert litres into millilitres, we multiply the number of litres by 1000.

$$\text{Thus, } 4 \text{ L} = (4 \times 1000) \text{ mL} = 4000 \text{ mL};$$

$$8 \text{ L} = (8 \times 1000) \text{ mL} = 8000 \text{ mL};$$

Rule 2: To convert litres and millilitres into millilitres, we multiply the number of litres by 1000 and add to it the number of millilitres.

$$\text{Thus, } 2 \text{ L } 426 \text{ mL} = (2 \times 1000) \text{ mL} + 426 \text{ mL}$$

$$= 2000 \text{ mL} + 426 \text{ mL} = 2426 \text{ mL}.$$

Example 1: Convert the following into millilitres.

- (a) 9 L (b) 1 L 625 mL (c) 7 L 65 mL (d) 6 L 6 mL

Solution: We have:

$$(a) \quad 9 \text{ L} = (9 \times 1000) \text{ mL} = 9000 \text{ mL}.$$

$$(b) \quad 1 \text{ L } 625 \text{ mL} = (1 \times 1000) \text{ mL} + 625 \text{ mL} = 1000 \text{ mL} + 625 \text{ mL} = 1625 \text{ mL}.$$

$$(c) \quad 7 \text{ L } 65 \text{ mL} = (7 \times 1000) \text{ mL} + 65 \text{ mL} = 7000 \text{ mL} + 65 \text{ mL} = 7065 \text{ mL}.$$

$$(d) \quad 6 \text{ L } 6 \text{ mL} = (6 \times 1000) \text{ mL} + 6 \text{ mL} = 6000 \text{ mL} + 6 \text{ mL} = 6006 \text{ mL}.$$

Example 2: Convert the following into litres and millilitres.

- (a) 5635 mL (b) 2084 mL (c) 4008 mL

Solution: We have:

$$(a) \quad 5635 \text{ mL} = 5000 \text{ mL} + 635 \text{ mL} = 5 \text{ L} + 635 \text{ mL} = 5 \text{ L } 635 \text{ mL}.$$

$$(b) \quad 2084 \text{ mL} = 2000 \text{ mL} + 84 \text{ mL} = 2 \text{ L} + 84 \text{ mL} = 2 \text{ L } 84 \text{ mL}.$$

$$(c) \quad 4008 \text{ mL} = 4000 \text{ mL} + 8 \text{ mL} = 4 \text{ L} + 8 \text{ mL} = 4 \text{ L } 8 \text{ mL}.$$



Exercise 52

1. Fill the appropriate unit (L or mL).

- (a) Water in a storage tank
- (b) A glass of milk
- (c) A bucket full of water
- (d) A sachet of ketchup
- (e) Medicine in a syringe
- (f) Petrol filled in a car
- (g) Nail polish in a bottle



2. Convert the following into millilitres.

- (a) 3 L (b) 6 L (c) 7 L 800 mL
(d) 2 L 764 mL (e) 1 L 80 mL (f) 5 L 5 mL

3. Convert the following into litres and millilitres.

- (a) 5000 mL (b) 8100 mL (c) 9372 mL
(d) 2048 mL (e) 6002 mL (f) 3020 mL

Addition and Subtraction of Capacities (or Volumes)

On the basis of the above principle, we have the following rules:

Rule 1: To add two or more capacities (or volumes), we simply add the numbers and put down the unit. Observe the following additions:

$\begin{array}{r} 73\text{ L} \\ + 59\text{ L} \\ \hline 132\text{ L} \end{array}$	$\begin{array}{r} 372\text{ L} \\ + 268\text{ L} \\ \hline 640\text{ L} \end{array}$	$\begin{array}{r} 28\text{ mL} \\ + 45\text{ mL} \\ \hline 73\text{ mL} \end{array}$	$\begin{array}{r} 56\text{ mL} \\ 674\text{ mL} \\ + 167\text{ mL} \\ \hline 897\text{ mL} \end{array}$
1 ← Carry	1 1 ← Carry	1 ← Carry	1 1 ← Carry

Rule 2: To subtract a given capacity (or volume) from the other, we simply subtract the numbers and put down the unit.

$\begin{array}{r} 88\text{ L} \\ - 68\text{ L} \\ \hline 20\text{ L} \end{array}$	$\begin{array}{r} 516\text{ mL} \\ - 378\text{ mL} \\ \hline 138\text{ mL} \end{array}$	$\begin{array}{r} 2910\text{ L} \\ - 315\text{ L} \\ \hline 2595\text{ L} \end{array}$
8 15	5 16 15	2 9 10

Addition of Measures in Litres and Millilitres

We arrange the given measures in two columns of L and mL and add them separately.

Here, we shall write mL as 3-digit numbers.

Thus, we write 75 mL as 075 mL and 5 mL as 005 mL.

The following examples will make the idea more clear.





Solved Examples

Example 1: Add 17 L 450 mL and 23 L 275 mL.

Solution: Arranging the given measures in two columns of L and mL, we get:

L	mL
① 1 7	① 4 5 0
+ 2 3	2 7 5
4 0	7 2 5

Step 1. Adding millilitres.

$$450 \text{ mL} + 275 \text{ mL} = 725 \text{ mL.}$$

Write 725 under mL column.

Step 2. Adding litres.

$$17 \text{ L} + 23 \text{ L} = 40 \text{ L.}$$

Write 40 under L column.

$$\therefore \text{Sum} = 40 \text{ L } 725 \text{ mL.}$$

Example 2: Add 24 L 978 mL and 48 L 396 mL.

Solution: Arranging the given measures in two columns of L and mL, we get:

L	mL
① ① 2 4	① ① 9 7 8
+ 4 8	3 9 6
7 3	3 7 4

Step 1. Adding millilitres.

$$978 \text{ mL} + 396 \text{ mL} = 1374 \text{ mL}$$

$$= 1000 \text{ mL} + 374 \text{ mL}$$

$$= 1 \text{ L} + 374 \text{ mL.}$$

Write 374 under mL column and carry over 1 to L column.

Step 2. Adding litres.

$$1 \text{ L (carried over)} + 24 \text{ L} + 48 \text{ L} = 73 \text{ L}$$

Write 73 under L column.

$$\therefore \text{Sum} = 73 \text{ L } 374 \text{ mL.}$$

Note : We may note here that addition may be performed as ordinary numbers provided, we write millilitres as 3-digit numbers.

Example 3: Add 45 L 738 mL, 37 L 604 mL and 19 L 85 mL.

Solution: Arranging the given measures in two columns of L and mL, we get

L	mL
② ① 4 5	① ① 7 3 8
3 7	6 0 4
+ 1 9	0 8 5
1 0 2	4 2 7

Step 1. Adding millilitres.

$$738 \text{ mL} + 604 \text{ mL} + 85 \text{ mL} = 1427 \text{ mL}$$

$$= 1000 \text{ mL} + 427 \text{ mL}$$

$$= 1 \text{ L} + 427 \text{ mL.}$$

Write 427 under mL column and carry over 1 to L column.

Step 2. Adding litres.

$$1 \text{ L (carried over)} + 45 \text{ L} + 37 \text{ L} + 19 \text{ L} = 102 \text{ L.}$$

Write 102 under L column.

$$\therefore \text{Sum} = 102 \text{ L } 427 \text{ mL.}$$



Exercise 53

Add:

1.

	2	3	8	L
+	3	5	6	L

2.

	5	7	8	L
+	1	8	4	L

3.

	4	8	5	L
+	2	6	7	L

4.

L	mL
8	6 4 5
+	3 2 5 5

5.

L	mL
9	7 0 6
+	1 1 2 7 9

6.

L	mL
3 2	9 4 8
+	9 0 5 2

7.

L	mL
1 7	8 2 6
+	2 3 5 7 8

8.

L	mL
4 6	9 5 0
+	3 9 4 7 5

9.

L	mL
7 6	5 8 6
+	4 5 6 7 4

10.

L	mL
9	6 8 5
	1 4 0 9 5
+	1 6 0 0 7

11.

L	mL
2 3	0 0 8
	4 7 0 5 6
+	8 7 9 5

12.

L	mL
2 6	9 0 0
	1 0 3 7 0 0
+	6 5 5 0 0

13.

L	mL
0	7 3 5
	1 6 0 8 5
+	2 0 5 9 8 5

14.

L	mL
2 0 5	8 5 3
	6 9 7 8 6
+	8 6 7 8

15.

L	mL
1 2 5	6 7 5
	6 5 8 5 0
+	0 6 5 5

Arrange in columns and add.

16. 51 L 700 mL and 48 L 300 mL

17. 18 L 564 mL, 37 L 658 mL and 9 L 75 mL

18. 9 L 9 mL, 18 L 18 mL, 125 L 125 mL and 730 mL

19. 75 L 75 mL, 7 L 7 mL, 65 L 915 mL and 1 L 545 mL



Subtraction of Measures in Litres and Millilitres

We arrange the given measures in two columns of L and mL and subtract them separately.
Be careful, write mL as 3-digit numbers.
Study the following examples.



Solved Examples

Example 1: Subtract 29 L 358 mL from 43 L 605 mL.

Solution: Arranging the given measures in columns of L and mL, we get :

L	mL
43	605
4 3	6 0 5
- 29	358
14	247

Step 1. Subtracting millilitres.

$$605 \text{ mL} - 358 \text{ mL} = 247 \text{ mL.}$$

Write 247 under mL column.

Step 2. Subtracting litres.

$$43 \text{ L} - 29 \text{ L} = 14 \text{ L.}$$

Write 14 under L column.

$\therefore$ Difference = 14 L 247 mL.



Example 2: Subtract 69 L 475 mL from 93 L 150 mL.

Solution: Arranging the given measures in columns of L and mL, we get.

L	mL
93	150
9 3	1 5 0
- 69	475
23	675

Step 1. Subtracting millilitres.

Since $475 > 150$, we cannot subtract 475 mL from 150 mL.

So, we borrow 1 L, leaving behind 92 L.

$$\text{Now, } 1 \text{ L } 150 \text{ mL} = 1000 \text{ mL} + 150 \text{ mL} = 1150 \text{ mL.}$$

$$\therefore 1150 \text{ mL} - 475 \text{ mL} = 675 \text{ mL.}$$

Step 2. Subtracting litres.

$$92 \text{ L} - 69 \text{ L} = 23 \text{ L.}$$

Write 23 under L column.

$\therefore$ Difference = 23 L 675 mL.

Example 3: Subtract 16 L 89 mL from 41 L 5 mL.

Solution: Arranging the given measures in columns of L and mL, we get:

L	mL
41	5
4 1	0 5
- 16	89
24	916

Step 1. Subtracting millilitres.

Since $89 > 5$, we cannot subtract 89 mL from 5 mL.

So, we borrow 1 L, leaving behind 40 L.

$$\text{Now, } 1 \text{ L } 5 \text{ mL} = 1000 \text{ mL} + 005 \text{ mL} = 1005 \text{ mL.}$$

$$\therefore 1005 \text{ mL} - 89 \text{ mL} = 916 \text{ mL.}$$

Write 916 under mL column.

Step 2. Subtracting litres.

$$40 \text{ L} - 16 \text{ L} = 24 \text{ L.}$$

Write 24 under L column.

$\therefore$ Difference = 24 L 916 mL.

Note: Please note that subtraction may be performed as in ordinary numbers provided, we write millilitres as 3-digit numbers.



Exercise 54

Subtract:

1.

	3	6	0	mL
-	2	7	5	mL

2.

	5	0	0	L
-	2	9	2	L

3.

	8	7	3	L
-	5	9	5	L

4.

	L		mL
	5	4	9 5 0
-	2	8	7 6 5

5.

	L		mL
	3	1	3 1 2
-	2	5	0 8 5

6.

	L		mL
	8	0	2 0 0
-	6	8	7 2 5

7.

	L		mL
	1	3	0 7 5 0
-	8	4	9 7 5

8.

	L		mL
	4	0	3 3 2 0
-	1	5	9 4 5 6

9.

	L		mL
	7	2	6 0 5
-	9	7	1 6

10.

	L		mL
	1	0	0 0 0
-	0	9	2 5

11.

	L		mL
	6	1	0 0 5
-	1	9	0 6 6

12.

	L		mL
	5	1	3 0 6 5
-	2	1	8 7 7 7

Arrange in columns and then subtract:

13. 784 mL from 3L

14. 23 L 425 mL from 50 L

15. 7 L 165 mL from 100 L 100 mL

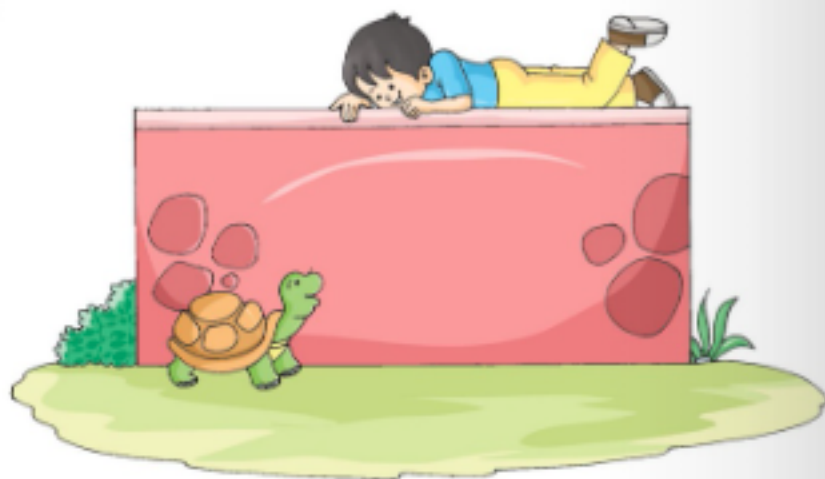
16. 78 L 385 mL from 97 L 30 mL

Find the difference between:

17. 4 L 575 mL and 9 L 55 mL

18. 34 L 75 mL and 101 L 60 mL

19. 2 L 525 mL and 3 L 100 mL



Word Problems

Example 1: A bucket holds 5 L 325 mL of milk. If 1 L 785 mL of water is added to it to make the bucket full, what is the capacity of the bucket?

Solution:

		L	mL
Quantity of milk in the bucket	=	5	3 2 5
Quantity of water added to it	=	+ 1	7 8 5
Total quantity of the mixture in the bucket	=	7	1 1 0

Hence, the capacity of the bucket is 7 L 110 mL.

Example 2: The capacity of petrol tank in a car is 26 L 200 mL. It contains 17 L 525 mL of petrol. How much more petrol it can take in?

Solution:

		L	mL
Capacity of the petrol tank in the car	=	2 6	2 0 0
Quantity of petrol contained in the car	=	- 1 7	5 2 5
Quantity of petrol that can be added to it	=	8	6 7 5

Hence, 8 L 675 mL of more petrol can be added to the tank.

Example 3: A big drum contained 105 litres of kerosene. During the day 68 L 752 mL of kerosene was sold and 587 mL was spilled. How much of the kerosene was left in the drum?

Solution:

		L	mL
Quantity of kerosene sold	=	6 8	7 2 5
Quantity of kerosene spilled	=	+ 0	5 8 7
Quantity of kerosene withdrawn from the drum	=	6 9	3 1 2

		L	mL
Now, total quantity of kerosene in the drum	=	1 0 5	0 0 0
Quantity of kerosene withdrawn from the drum	=	- 6 9	3 1 2
Quantity of kerosene left in the drum	=	3 5	6 8 8

Hence, the quantity of kerosene left in the drum is 35 L 688 mL.



Exercise 55

1. On Krishna's birthday her mother bought 6 L 750 mL of milk from one dairy and 5 L 250 mL from another dairy. How much of milk did she buy on that day?
2. A bucket contains 4 L 975 mL of milk. If 1 L 50 mL of water is added to it, what quantity of mixture will be there in the bucket?

3. Three buckets containing 8 L 125 mL, 5 L 850 mL and 4 L 60 mL of milk respectively are inverted in an empty drum. Find the total quantity of milk in the drum.
4. A tin contained 15 L 320 mL of oil. Out of it, 875 mL is wasted due to a leak in the tin. How much oil is left in the tin?
5. A bucket has a capacity of 6 litres. If it contains 2 L 325 mL of water, how much more water can be poured into it?
6. Reena takes 765 mL of cold drink from a jug containing 1 L 300 mL. How much cold drink is left in the jug?
7. Before starting for a journey, the tank of a car was filled with 25 L of petrol. At the end of the journey, 8 L 350 mL of petrol was left in the car. How much petrol was used in the journey?
8. Pamila was carrying 4 L 250 mL of milk in a can from a dairy to her home. On the way, she stumbled and some milk was spilled. On reaching home, she found that 3 L 475 mL of milk is left in the can. How much of milk was spilled on the way?
9. A shopkeeper had 60 litres of refined oil. He sold 12 L 50 mL to one customer, 20 L 550 mL to another customer and 14 L 400 mL to a third customer. How much oil is left with him?
10. A new tank was installed on a petrol pump. It had a capacity of 1000 litres. Two tankers poured 360 L 850 mL and 385 L 350 mL of petrol into it. How much more petrol can the tank hold?
11. Sunita bought 8 L 500 mL of milk. Out of it, 4 L 225 mL was used in preparing tea for a party, 2 L 50 mL was used for preparing cheese and 899 mL was taken by her child. She used the remaining quantity of milk for preparing curd. How much milk was used for curd?



Things to Remember

1. The quantity of liquid that a container or vessel can hold is called its capacity.
2. The standard unit of capacity is litre.
3. 1 litre = 1000 mL.
4. To add the measures in litres and millilitres, we arrange them in two columns of L and mL and add them separately.
5. To find the difference between two given measures (in litres and millilitres), we arrange them in two columns of L and mL and subtract them separately.
6. While adding or subtracting, we always write mL as 3-digit numbers.

Thus, we write 5 mL as 005 mL; 65 mL as 065 mL, etc.

**QUESTION BAG 1****(Objective Type Questions)****Tick (✓) the correct answer.**

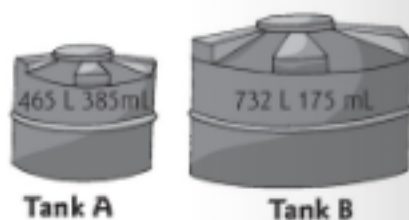
1. The capacity of a tablespoon is measured in
(a) millilitres ☐ (b) litres ☐ (c) gallons ☐ (d) grams ☐
2. Arrange the following in the order of increasing capacities.
1. Glass 2. Jug 3. Cup 4. Bucket
(a) 2, 1, 3, 4 ☐ (b) 3, 1, 2, 4 ☐ (c) 4, 2, 1, 3 ☐ (d) 1, 3, 2, 4 ☐
3.  +  +  = 750 mL. Then,  =
(a) 200 mL ☐ (b) 250 mL ☐ (c) 300 mL ☐ (d) None of these ☐
4. $2\text{ L } 350\text{ mL} - 750\text{ mL} = ?$
(a) 1 L 500 mL ☐ (b) 1 L 600 mL ☐ (c) 1 L 650 mL ☐ (d) 1 L 700 mL ☐
5. $6\text{ L } 650\text{ mL} + 18\text{ L } 750\text{ mL} = ?$
(a) 24 L 400 mL ☐ (b) 25 L 300 mL ☐ (c) 25 L 400 mL ☐ (d) 26 L 300 mL ☐
6. If 2 jugs can hold 8 litres of water, then 5 jugs can hold..... litres.
(a) 16 ☐ (b) 18 ☐ (c) 20 ☐ (d) 24 ☐
7. A jug of capacity 2 litres can be half-filled by pouring 5 full mugs of water into it. How much water does the mug hold when it is full?
(a) 200 mL ☐ (b) 250 mL ☐ (c) 400 mL ☐ (d) 500 mL ☐
8. $50\text{ L } 250\text{ mL} - 18\text{ L } 500\text{ mL} = ?$
(a) 30 L 750 mL ☐ (b) 31 L 250 mL ☐ (c) 31 L 750 mL ☐ (d) 32 L 250 mL ☐
9. 3 glasses of water is equal to 1 litre. If a bucket can hold 8 litres of water, how many glasses of water is required to fill it?
(a) 9 ☐ (b) 16 ☐ (c) 24 ☐ (d) 25 ☐
10. Reena purchased 5 L 250 mL of milk. She used 3 L 750 mL during the day. What quantity of milk is she left with?
(a) 1250 mL ☐ (b) 1400 mL ☐ (c) 1500 mL ☐ (d) 1550 mL ☐

11. An oil drum has 42 L 800 mL of oil in it. If 14 L 900 mL of oil is taken out of it, how much is left?

- (a) 27 L 100 mL ☐ (b) 27 L 900 mL ☐ (c) 28 L 100 mL ☐ (d) 28 L 900 mL ☐

12. Two water tanks are shown below. Tank B holds how much water more than Tank A?

- (a) 266 L 790 mL ☐ (b) 267 L 365 mL ☐
(c) 267 L 560 mL ☐ (d) 333 L 590 mL ☐



QUESTION BAG 2

1. Fill in the blank boxes.

- (a) 250 mL + mL = 1 L
(b) 200 mL + 500 mL + mL = 1 L
(c) mL + 600 mL = 1 L
(d) 350 mL + mL + 200 mL = 1 L
(e) 1 L + mL + 500 mL = 2 L

2. Put the correct symbol >, < or = in the boxes.

- (a) 5 L 25 mL 5225 mL (b) 2 L 822 mL 2822 mL
(c) 7 L 500 mL 7050 mL (d) 3 L 810 mL 3081 mL
(e) 9 L 85 mL 9850 mL

3. A tanker had 800 litres of petrol. Out of it, 608 L 500 mL was delivered to a petrol pump. How much petrol is left in the tanker now?
4. There was 62 L 320 mL water in a drum. Deepak took out 9 L 535 mL water in a bucket. How much water is left in the drum?
5. Nisha went to her school with a bottle having 2 litres of water in it. She drank 230 mL of water before the break, 365 mL during the break and 225 mL while returning home. How much water is left in the bottle?
6. Amit mixed 6 L 885 mL of hot water with 5 L 675 mL of cold water in the bucket. How much water is there in the bucket?
7. A village has two wells. One has 367 L 588 mL of water and the other has 496 L 747 mL of water. How much water do the two wells have in all?

